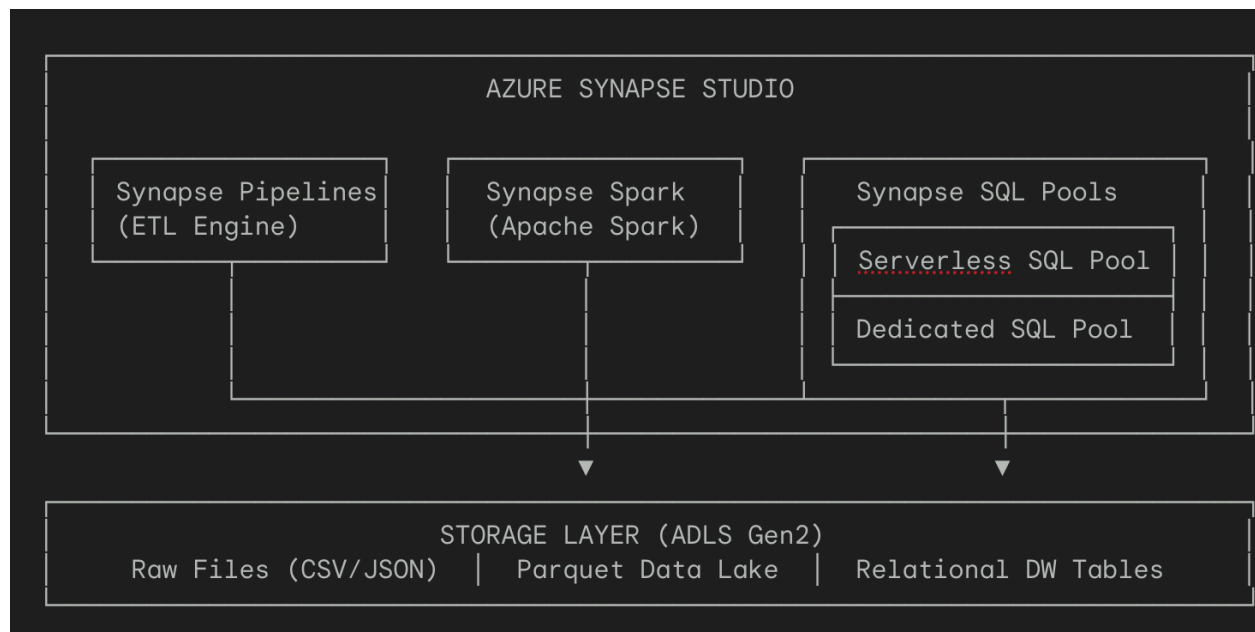


Azure Synapse Analytics vs. Azure Databricks Architecture

1. Architectural Overview

Azure Synapse Analytics Architecture

Azure Synapse Analytics is a unified enterprise analytics service that brings together data integration, enterprise data warehousing, and big data processing into a single interface (Synapse Studio).

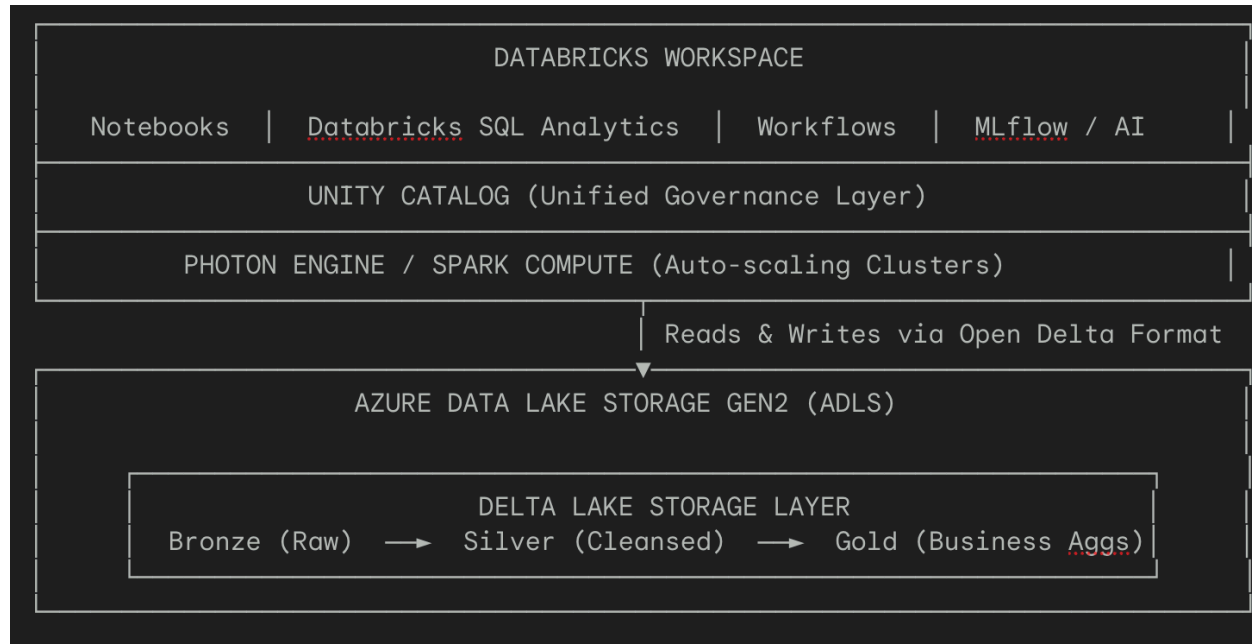


Core Components

1. **Dedicated SQL Pools (formerly Azure SQL DW):** Provisioned Massively Parallel Processing (MPP) architecture for high-performance enterprise data warehousing.
2. **Serverless SQL Pools:** On-demand query engine designed to execute ad-hoc T-SQL queries over raw files stored in ADLS Gen2 without cluster provisioning.
3. **Synapse Spark Pools:** Managed Apache Spark environment integrated directly into Synapse Studio for Python, Scala, and PySpark workflows.
4. **Synapse Pipelines:** Native ETL/ELT orchestration engine built on the same core underlying technology as Azure Data Factory (ADF).

Azure Databricks Architecture

The modern Azure Lakehouse architecture decouples compute from storage, using **Delta Lake** as an open ACID table format directly on top of Azure Data Lake Storage Gen2 (ADLS Gen2).

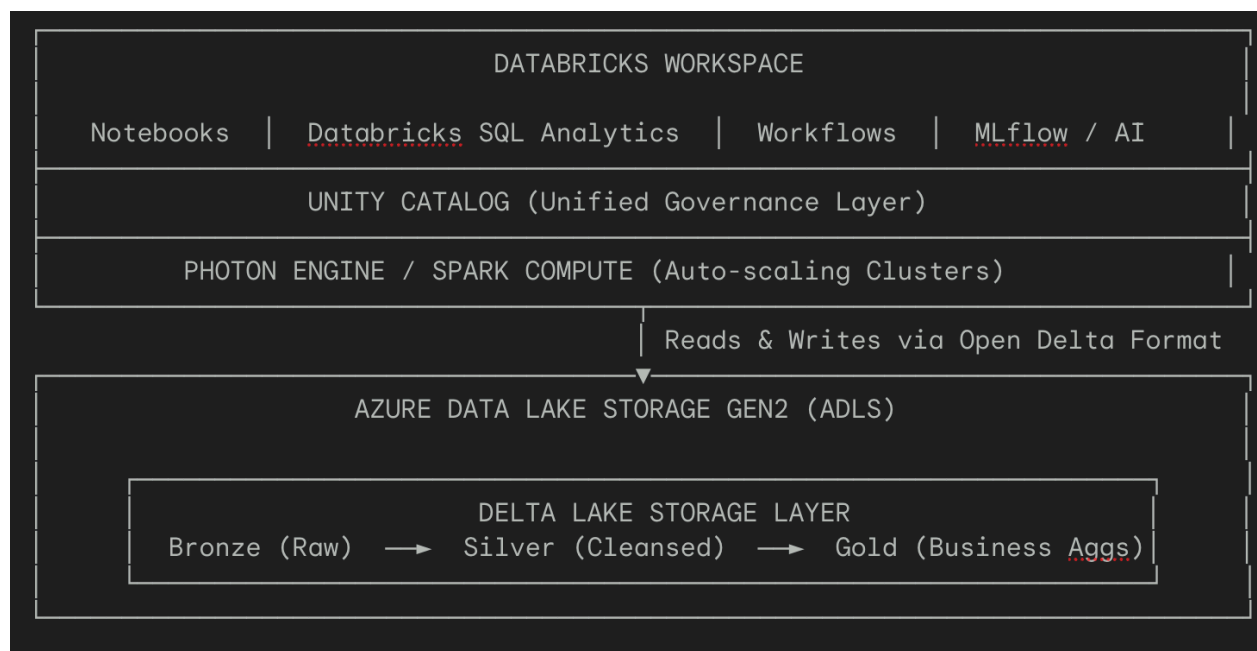


Core Components

1. **Control Plane & Data Plane Isolation:** Control Plane (UI, metadata) is managed by Databricks; Data Plane (Worker/Driver VMs) runs directly inside the customer's Azure Virtual Network.
2. **Photon Engine:** A vectorized C++ query processing engine built to accelerate Spark SQL and DataFrame workloads.
3. **Unity Catalog:** A centralized governance solution across all workspaces, providing fine-grained access control (column/row level) for data assets and AI models.
4. **Delta Lake:** Open ACID transaction log layer over Parquet files in ADLS Gen2, providing time travel, schema enforcement, and concurrent mutation operations (MERGE, UPDATE, DELETE).

2. Deep Dive: Synapse Dedicated SQL Pool Architecture

Synapse Dedicated SQL Pools execute queries using a **Massively Parallel Processing (MPP)** architecture. Compute resources are measured in **Data Warehouse Units (DWUs)**.



Operational Mechanics

1. **Control Node:** Accepts client connections, parses T-SQL code, optimizes execution plans, and coordinates parallel execution across compute nodes.
2. **Compute Nodes:** Worker units that run SQL execution threads on allocated portions of the dataset. Higher DWU levels scale up the number of active Compute Nodes.
3. **60 Storage Distributions:** Behind the scenes, every Dedicated SQL Pool splits data across exactly **60 fixed storage distributions**.
4. **Data Movement Service (DMS):** Internal transport channel that moves data across compute nodes during execution to complete joins and aggregations.

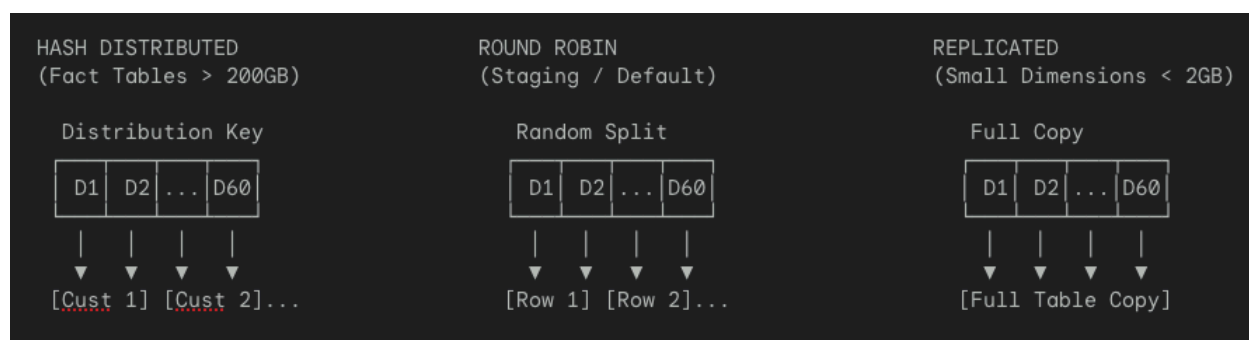
Table Distribution Strategies

To optimize query latency in Dedicated SQL Pools, tables must be designed using one of three distribution models:

HASH DISTRIBUTED
(Fact Tables > 200GB)

ROUND ROBIN
(Staging / Default)

REPLICATED
(Small Dimensions < 2GB)



1. **Hash Distribution:** Uses a hash function on a single chosen column to map rows evenly across the 60 distributions. Optimal for large fact tables (>200GB).
2. **Round-Robin Distribution:** Distributes rows evenly in a sequential sequence without checking column values. Ideal for temporary staging tables due to fast loading speeds.
3. **Replicated Tables:** Maintains a full copy of the entire table on every compute node. Eliminates data shuffling during joins on smaller dimension tables (2GB).

3. Deep Dive: Synapse Serverless SQL vs. Databricks SQL

Feature	Azure Synapse Serverless SQL Pool	Databricks SQL (Serverless SQL Warehouse)
Pricing Model	Charged per TB of data scanned	Charged based on DBUs (Databricks Units) and compute runtime (typically billed per minute)
Cold Start Latency	Near zero (queries can be submitted immediately)	Very low (typically a few seconds for serverless warehouses to start)
Primary Data Source	CSV, JSON, Parquet, Delta files stored in Azure Data Lake Storage (ADLS)	Delta Lake tables managed through Unity Catalog (can also query other supported formats)

Best Used For	Ad-hoc data exploration, querying raw files, lightweight reporting, and creating logical views	High-performance BI dashboards (Power BI, Tableau), interactive analytics, and production SQL workloads
Optimization Focus	Partition pruning (folder-based partitioning), efficient file formats (Parquet/Delta)	Photon Engine, Caching, Z-Ordering, Liquid Clustering, Query Optimization, Predictive Optimization

4. Key Architectural Trade-offs & Nuances

1. Data Storage: Open Format vs. Proprietary Relational

- **Databricks Lakehouse:** Stores data in open format (**Delta / Parquet**) directly in your ADLS Gen2 storage account. Avoids vendor lock-in; any processing engine (Spark, Presto, DuckDB, Synapse Serverless) can read the underlying files.
- **Synapse Dedicated SQL:** Ingests data into internal relational database storage structures mapped to 60 distributions. Unlocking data requires exporting via CETAS (Create External Table As Select) or PolyBase/Copy statements.

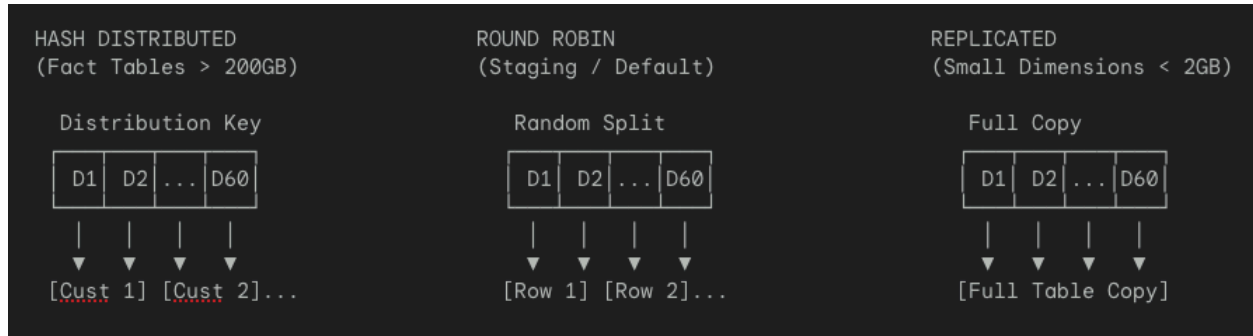
2. Compute Scaling Dynamics

- **Databricks:** Compute clusters scale out (adding worker nodes) and scale up dynamically based on queue depth and workload demand, with auto-termination when idle.
- **Synapse Dedicated SQL:** Compute scales by adjusting DWU levels. Scaling involves a brief pause in query execution while underlying compute nodes are reallocated over the 60 distributions.

3. Orchestration: ADF vs. Synapse Pipelines vs. Workflows

- **Azure Data Factory / Synapse Pipelines:** Control-flow orchestrators ideal for enterprise-wide pipeline management, on-premises SHIR connectivity, and low-code integrations.
- **Databricks Workflows:** Task orchestrator native to the lakehouse platform, designed to execute notebooks, Python scripts, DLT pipelines, and SQL queries with low operational cost using ephemeral job clusters.

5. Decision Framework: When to Choose Which Architecture?



Choose Azure Synapse Analytics when:

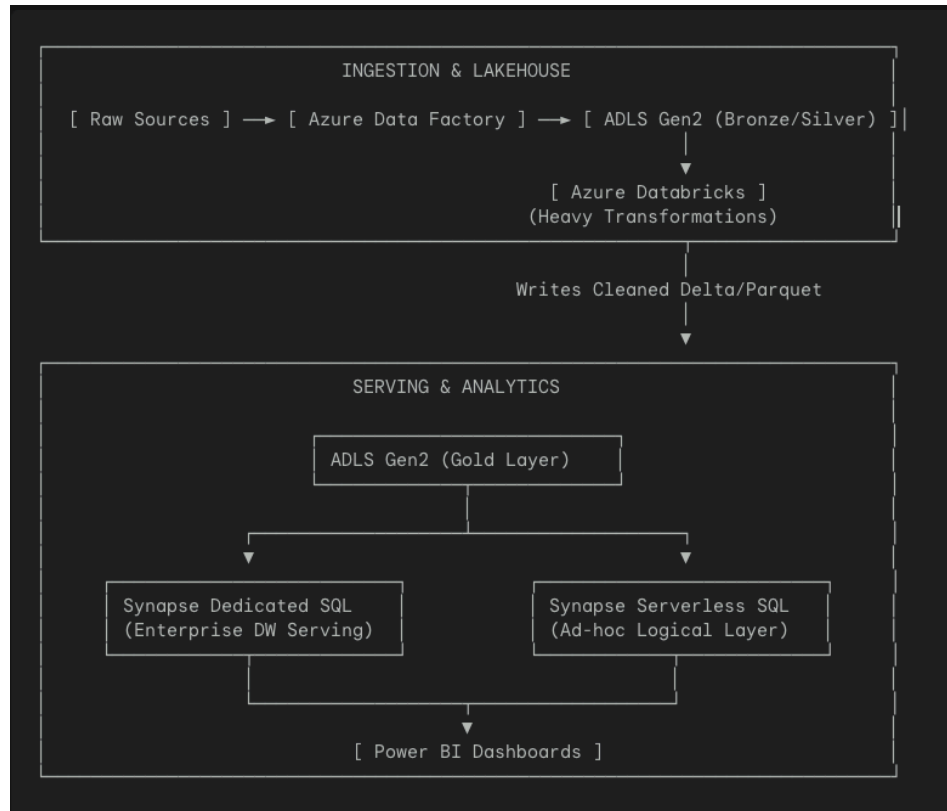
1. Your team consists primarily of SQL developers migration-bound from traditional SQL Server / Teradata / Netezza platforms.
2. The primary requirement is fixed-capacity enterprise data warehousing with predictable month-over-month costs .
3. You need simple, out-of-the-box integration between data ingestion (Pipelines) and data warehousing without managing external Spark platforms.

Choose Azure Databricks when:

1. You are building an open **Medallion Architecture** (Bronze → Silver → Gold) using Delta Lake.
2. Your platform requires advanced Python/PySpark, machine learning (MLflow), streaming (Structured Streaming), or AI workloads.
3. You prefer decoupled storage to eliminate vendor lock-in, enabling fine-grained governance via Unity Catalog.

6. Enterprise Hybrid Architecture Pattern

In enterprise environments, organizations frequently combine both tools into a unified pipeline:



In this hybrid model:

1. **Azure Data Factory** handles raw data ingestion from legacy systems into ADLS Gen2.
2. **Azure Databricks** performs complex distributed transformations, cleansing, and aggregation, writing output directly to Delta format in the Gold storage container.
3. **Azure Synapse**

(Dedicated SQL / Serverless SQL) serves aggregated datasets to corporate BI tools and business users who require standard T-SQL endpoints.